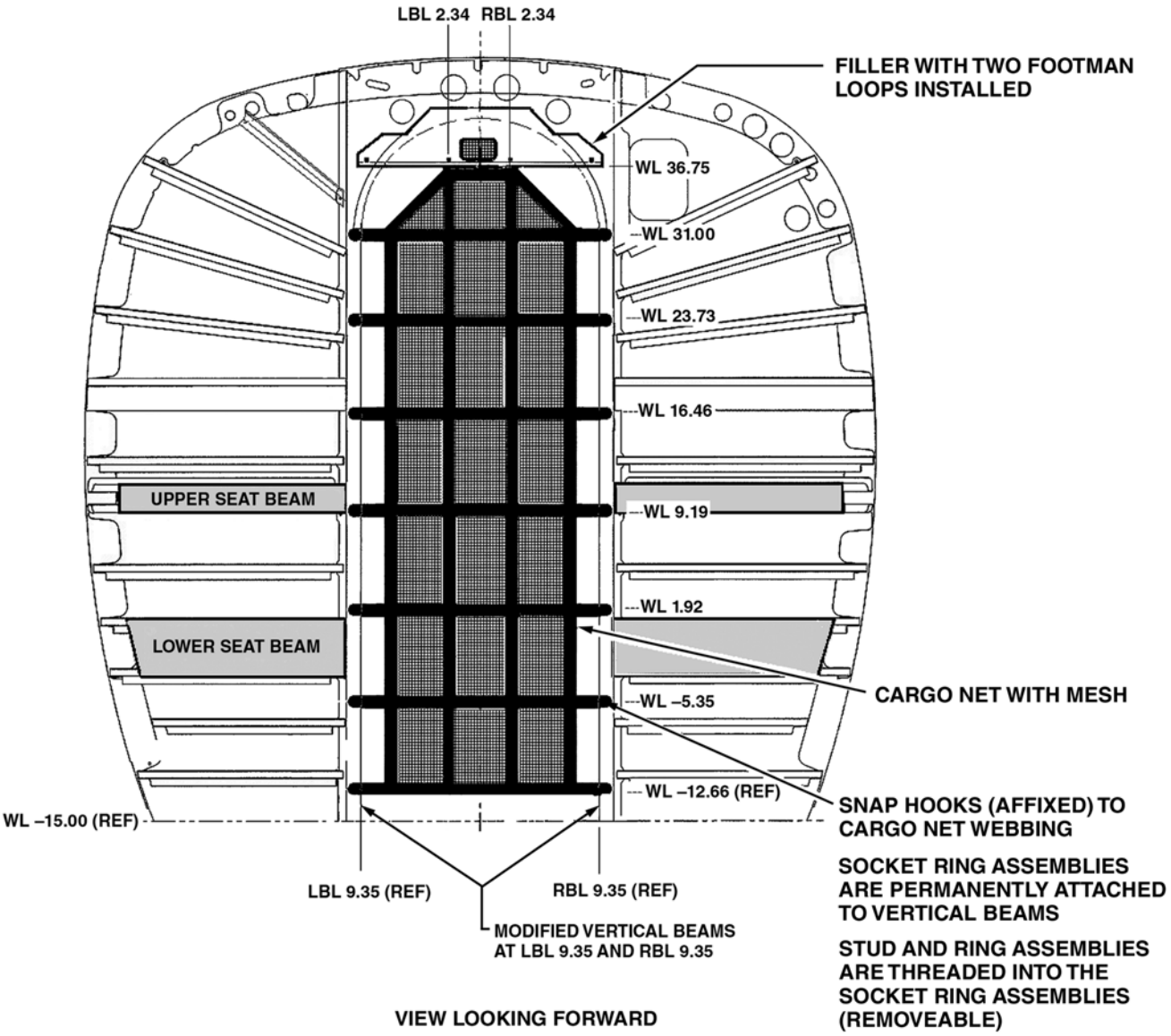
STA 332 Modified Bulkhead and Cargo Net Installation – General Data

1. General

The STA 332 bulkhead is modified to provide an opening into the rear baggage compartment as shown in Figure 1.

A removable lightweight cargo net is installed in the opening of the STA 332 modified bulkhead to separate the passenger cabin from the rear baggage compartment, and allows access to the rear baggage compartment from inside the aircraft.

The cargo net can be safely stowed in a Filler at the top of STA 332 bulkhead.



STA 332 Bulkhead Modification and Cargo Net Installation
Figure 1

2. Description

A. STA 332 Modified Bulkhead

The STA 332 bulkhead is modified by removing the center portion of the upper and lower seat beams and removing the center single folding seat. The two vertical beams are re-inforced to provide additional strength, and have stud and ring hardware installed to accommodate installation/removal of the cargo net.

B. Cargo Net

The cargo net is a light-weight web and mesh design which is secured in place by using fourteen snap hooks (seven per side), which are permanently sewn to the webbing. The cargo net snap hooks are “snapped” directly to the removable stud and ring assemblies on the vertical beams.

The top of the cargo net is secured using two Footman Loops attached to the filler installed at the top of the STA 332 bulkhead. The filler provides secure stowage when the cargo net is not in use.

C. Socket Ring Assemblies

The socket ring assemblies are permanently mounted to the modified vertical beams (7 socket ring assemblies per beam) and are threaded with a helicoil insert to accept the stud and ring assemblies.

D. Stud and Ring Assemblies

The stud and ring assemblies are removable and consist of a ring, which loops through a threaded stud. The cargo net snap hooks are “snapped” directly to the stud and ring assemblies.

3. Removal/Installation

A. Cargo Net

1) Removal

a) Open the clasp of each snap hook and disengage the snap hook from the stud and ring assembly down each side of the vertical beams (14 in total).

b) Prepare the cargo net for stowage inside the aircraft as per Para 3.A.3).

2) Installation

a) Prior to installing the cargo net, carry out the following:

1 Inspect the cargo net webbing for condition (refer to Para 4.). If damaged, repair or replace (refer to Para 5.).

2 Inspect the stud and ring assemblies on the aft side of the vertical beams for condition and corrosion. If damaged or missing, replace (refer to Para 5.).

3 Ensure that the identification tag of the cargo net faces aft.

b) Secure the top of the cargo net using two Footman Loops attached to the filler, located at the top of the STA 332 bulkhead.

c) Secure the cargo net in the opening of the STA 332 bulkhead by opening the clasp of each snap hook (14 in total) and engaging the snap hook into the stud and ring assemblies down each side of the vertical beams.

3) Stowage

a) Open the clasp of each snap hook (14 in total) and disengage the snap hook from the stud and ring assembly down each side of the bulkhead.

b) Roll the cargo net forward and upwards starting from the bottom.

c) Secure the rolled up cargo net in place using the two straps (Velcro) attached to the upper portion of the cargo net.

d) Ensure the cargo net is securely stowed.

B. Stud and Ring Assemblies

1) Removal

a) Remove the threaded stud and ring assemblies by turning the threaded assembly counterclockwise until it disengages from the socket ring assembly.

2) Installation

a) The stud and ring assemblies shall be oriented in the horizontal position for proper installation of the cargo net. The stud may be shimmed by adding AN960JD416 washers (or equivalent), as required to obtain the necessary horizontal orientation.

Note Torque for installing the stud and ring assembly (1/4 x 28 UNF thread): Hand tight.

C. Socket Ring Assemblies

If removal of the socket ring assemblies is required from inside the vertical beams, contact Viking Air Limited.

4. Inspection

A. Cargo Net Mesh

1) Mesh Damage Limitations are as follows:

a) Damage consisting of tears and/or wear that exceed 25% of the total number of meshed panels, is considered to be non-serviceable and the net shall be replaced.

b) No more than two (2) patches per section is permitted for field repairs.

c) Complete panel replacement shall be done by Cargo Systems Inc. only.

d) Material used for all patches is to be provided by Cargo Systems Inc. (P/N A4828–0705), and shall be made in accordance with procedures and specifications prescribed by Cargo Systems Inc.

e) For tears that are close to the edge of the webbing, patches may overlap webbing.

f) All patches are to be installed on the side OPPOSITE (backside) of the webbing.

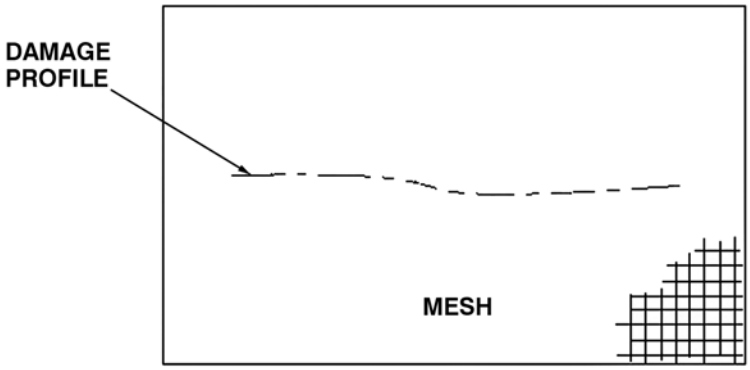
2) Mesh with Linear (Straight) Tears

Refer to Figure 2 for details.

a) Linear (straight) tears less than or equal to 2.0” long are acceptable. Repair NOT necessary at this time.

b) Field patch repair is acceptable for linear (straight) tears greater than 2.0” long, but less than 5.0” long. Refer to Figure 4.

c) Field patch repairs are not authorized for linear (straight) tears greater than 5.0” long. Return the cargo net to Cargo Systems Inc. for repair/replacement.



Mesh with Linear (Straight) Tears
Figure 2

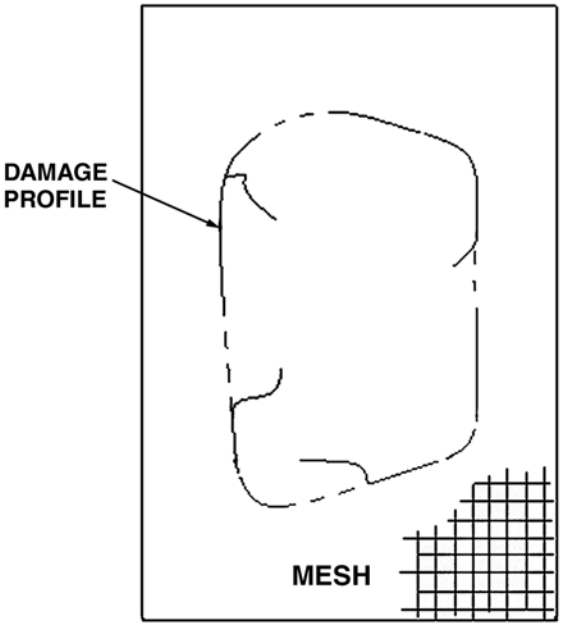
3) Mesh with Non-Linear (Uneven) Tears

Refer to Figure 3 for details.

a) Holes less than 1.0” diameter are acceptable. Repair is NOT necessary at this time.

b) Field patch repair is acceptable for holes greater than 1.0” diameter, but less than 3.0” diameter. Refer to Figure 5.

c) Field patch repairs are not authorized for tears greater than 3.0” diameter. Return the cargo net to Cargo Systems Inc. for repair/replacement.



Mesh with Non-Linear (Uneven) Tears
Figure 3

B. Cargo Net Webbing

1) Webbing Damage Limitations

a) Cargo net webbing that has tears, cuts and/or frayed edges (where the thread yarns have broken through) is not airworthy and must be repaired or returned to Cargo Systems Inc.

b) Light fraying (or fuzzing) of the webbing edges (where the thread yarns are still intact) is acceptable, provided that no tears or cuts are evident in the webbing yarns.

c) Replace the cargo net if any of the following conditions exits:

1 The length of an edge cut to the webbing exceeds the webbing thickness.

2 The penetration of the abrasion exceeds 15% of the webbing thickness.

3 If abrasion occurs on both sides of the webbing and the sum of the abrasion on both sides of the webbing exceeds 15% of the total webbing thickness.

4 Broken or worn stitching.

5 The identification tag is unreadable or missing.

C. Hardware

1) Snap Hooks

a) Inspect the snap hooks for security and condition. If corrosion is evident and cannot be removed using superfine steel wool or metal polish, the net should be removed from service and returned to Cargo Systems Inc. for repair.

b) Inspect the spring-loaded clasp to ensure that it is functional and operating smoothly.

c) Replace the snap hook if cracked, distorted or broken (refer to Para 5.D) or return the cargo net to Cargo Systems Inc. for snap hook replacement.

2) Socket Ring Assemblies (installed inside each beam)

a) Inspect the MS21209F4–15 helicoil inside each socket ring assembly for damage or corrosion. Remove and replace the helicoil, if necessary.

b) If removal of the socket ring assemblies from inside the vertical beam is required, contact Viking Air Limited.

3) Stud and Ring Assemblies

a) Inspect the stud and ring assemblies for security and condition.

b) Inspect the stud and ring assemblies for corrosion, distortion and function.

c) Ensure the ring can rotate freely within the stud and is mounted in the horizontal direction so the ring hangs down.

d) Replace stud and ring assemblies if cracked, distorted or broken.

D. STA 332 Modified Bulkhead

1) Vertical Beams

a) Inspect the aft strap doublers on both vertical beams for condition and wear at each of the stud and ring assembly locations.

b) Ensure the aft strap doublers are not gouged or distorted from contact with the rings from the cargo net stud and ring assembly.

Note If there's any damage to the aft strap doublers of the vertical beams, contact Viking Air Limited.

c) Inspect the vertical beam gussets (which attach the vertical beams to the STA 332 bulkhead stiffeners) for distortion and/or loose rivets. If there is any damage to the gussets of the vertical beams, contact Viking Air Limited.

5. Repairs

Note All repairs MUST always be accomplished by a qualified and approved repair facility.

A. Mesh Repair Materials (available from Cargo Systems Inc.)

- Thread: CSI P/N 29520–BLK, 29523–BLK if single layered mesh; or thick 29525–BLK if double layer mesh is used or overlapping webbing
- Stitching: 6 - 8 stitches per inch per MIL–STD-751, Type 301 Straight Stitch
- Mesh: CSI P/N A48280705 (Double layer recommended for durability; folded edge not required)

Note Material substitutions for the mesh and thread are not permitted unless approved by Cargo Systems Inc.

B. Cargo Net Mesh Repair

Cargo net mesh repair is possible providing:

1) Complete mesh panel replacement is supplied by Cargo Systems Inc. only.

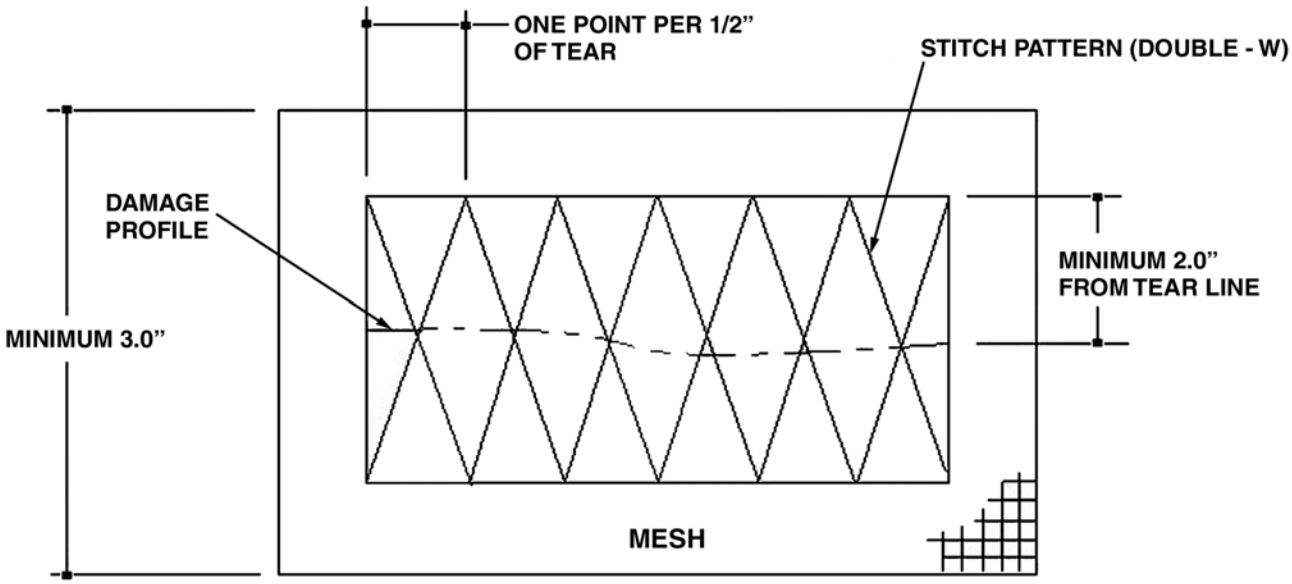
2) All mesh repair materials (P/N A48280705) are obtained from Cargo Systems Inc.

3) For tears that are in close proximity to the edge of the webbing, mesh patches may overlap webbing.

4) All mesh patches are installed on the side OPPOSITE (backside) of the webbing.

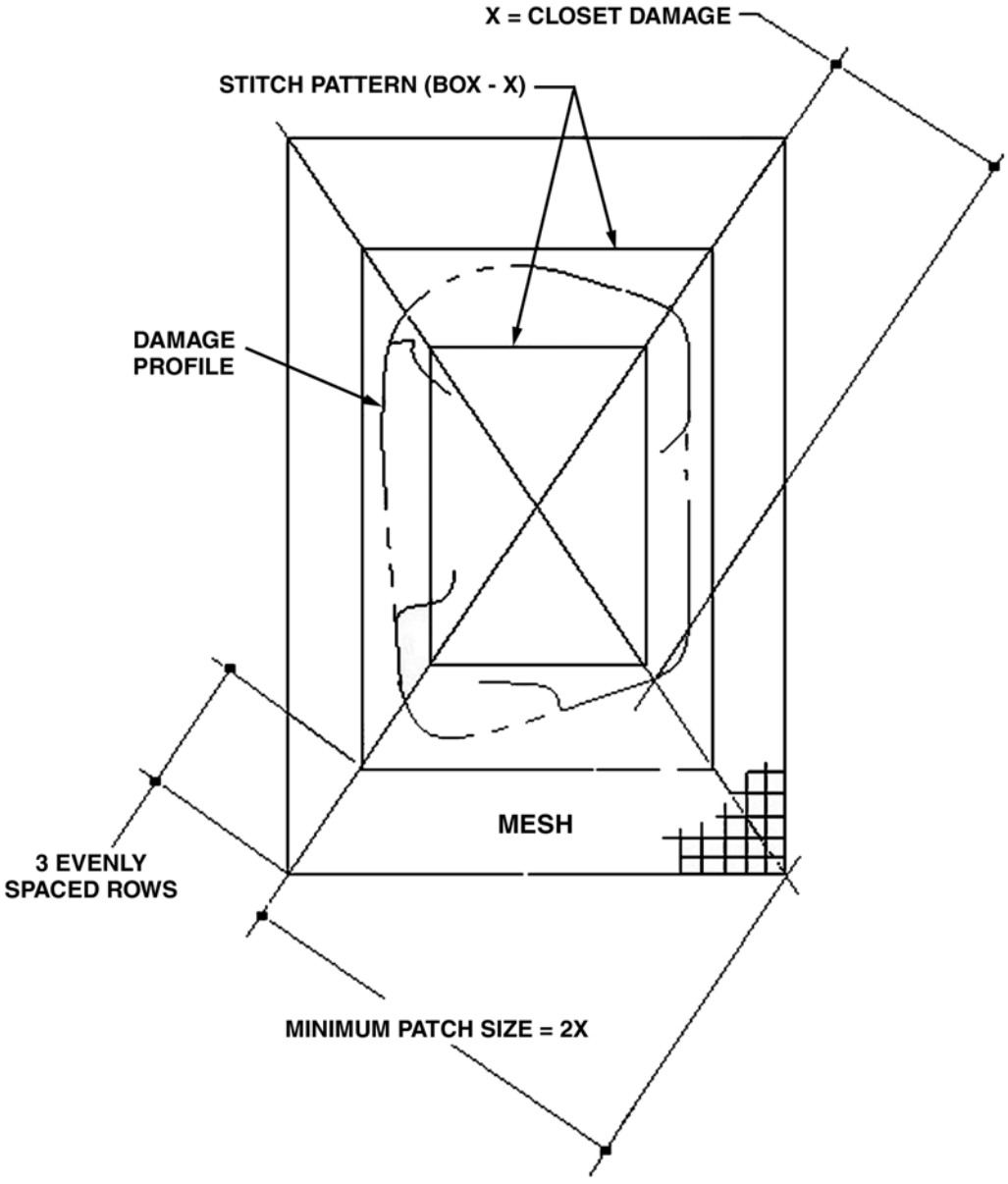
5) No more than two field repair patches are permitted for each meshed panel.

6) Repairs are made in accordance with procedures and specifications prescribed by Cargo Systems Inc.



Typical Stitch Pattern Layout (Double – W) for Mesh Repair with Straight (Linear) Tears
Figure 4

Note A double layer of mesh is recommended for this repair, using thicker 29525 BLK thread available from Cargo Systems Inc.



Typical Stitch Pattern Layout (Box – X) for Mesh Repair with Non-Linear (Uneven) Tears
Figure 5

Note A double layer of mesh is recommended for this repair, using thicker 29525 BLK thread available from Cargo Systems Inc.

C. Cargo Net Webbing Repair

Note Material substitutions for the webbing and thread are not permitted unless approved by Cargo Systems Inc.

1) Loose Stitching

Stitch patterns in the webbing that exhibit broken threads may be sewn over with bonded nylon thread (available from Cargo Systems Inc.) using a minimum 4 - 6 stitches per inch per MIL–STD–751, Type 301, straight stitch. Replacement thread (P/N 29525–BLK) is available from Cargo Systems Inc.

2) Cut/Frayed Webbing

In cases where the webbing is completely cut through or when the yarn threads have been severed, the webbing may be repaired by splicing, using a piece of MIL-Spec webbing (same as existing webbing) available from Cargo Systems Inc. The webbing splice shall extend 6.0” past the tear and shall have a minimum of 5.0” of stitching (in a 4 point Double W pattern) joining the web splice to the webbing on each side of the cut or the tear. Use a minimum of 70 stitches per pattern on either side of the cut. Stitches shall be 4 - 6 stitches per inch.

Note There may be only one webbing repair splice per webbing member (one length of load bearing webbing).

D. Hardware

1) Other Attachment Hardware

Check all hardware for security and condition and replace with same, if required.

E. STA 332 Modified Bulkhead

Repairs to the STA 332 bulkhead and vertical beam are NOT permitted unless approved by Viking Air Limited.

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